Dependent random choice

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Section 1.7 of *Graph Theory and Additive Combinatorics*, author manuscript updated 2024-06-18

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Source and scope. This is Theorem 1.7.5 of Yufei Zhao, *Graph Theory and Additive Combinatorics*, author manuscript updated 2024-06-18, printed pages 34–35 (physical pages 50–51 of the author PDF), together with its printed proof and the printed remark on parameters that stands between the statement and the proof. The statement and the proof are reproduced in full; nothing is added, shortened or repaired.

The theorem is stated inside Section 1.7, which introduces the dependent random choice technique. The one schematic in the proof is redrawn in TEX from the printed drawing: hosts $u_1, \dots, u_t$ on top, every displayed host joined to every displayed vertex of the common neighbourhood A below, with A shown as a shaded strip. The drawing is schematic in the source as well, and no additional hypothesis is read from it.

One printed detail is kept literally rather than normalised: the convexity display prints the averaging sum with the subscript $v \in V$ (not $v \in V(G)$). It is reproduced as printed and is recorded in the transcription fidelity review.

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Theorem 1.7.5 (Dependent random choice)**.** *Let n, r, m, t be positive integers and $\alpha > 0$. Then every graph G with n vertices and at least $\alpha n^2/2$ edges contains a vertex subset U with*

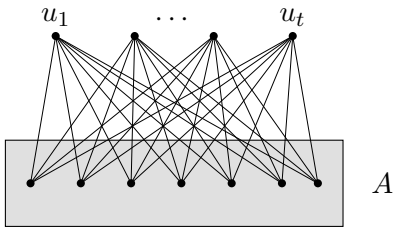
$$|U| \ \geq \ n\alpha^t - \binom{n}{r}\left(\frac{m}{n}\right)^t$$

such that every r-element subset S of U has more than m common neighbors in G.

Remark 1.7.6 (Parameters). In the theorem statement, t is an auxiliary parameter that does not appear in the conclusion. While one can optimize for t, it is instructive and convenient to leave it as is. The theorem is generally applied to graphs with at least n^{2-c} edges, for some small $c > 0$, and we can play with the parameters to get $|U|$ and m both large as desired.

Proof. We say that an r-element subset of $V(G)$ is "bad" if it has at most m common neighbors in G.

Let $u_1, \dots, u_t$ be vertices chosen uniformly and independently at random from $V(G)$. (These vertices are chosen "with replacement," i.e., they can repeat.) Let A be their common neighborhood. (Keep in mind that $u_1, \dots, u_t, A$ are random. It may be a bit confusing in this proof to see what is random and what is not.)



Each fixed vertex $v \in V(G)$ has probability $(\deg(v)/n)^t$ of being adjacent to all of $u_1, \ldots, u_t$, and so, by linearity of expectations and convexity,

$$\mathbb{E}|A| = \sum_{v \in V(G)} \mathbb{P}(v \in A) = \sum_{v \in V(G)} \Big(\frac{\deg(v)}{n}\Big)^t \;\geq\; n\Big(\frac{1}{n}\sum_{v \in V} \frac{\deg(v)}{n}\Big)^t \;\geq\; n\alpha^t.$$

For any fixed $R \subseteq V(G)$,

$$\mathbb{P}(R \subseteq A) = \mathbb{P}(R \text{ is complete to } u_1, \ldots, u_t) = \Big(\frac{\#\text{ common neighbors of } R}{n}\Big)^t.$$

If R is a bad r-vertex subset, then it has at most m common neighbors, and so

$$\mathbb{P}(R \subseteq A) \;\leq\; \Big(\frac{m}{n}\Big)^t.$$

Therefore, summing over all $\binom{n}{r}$ possible r-vertex subsets $R \subseteq V(G)$, by linearity of expectation,

$$\mathbb{E}[\text{the number of bad } r\text{-vertex subsets of } A] \;\leq\; \binom{n}{r}\Big(\frac{m}{n}\Big)^t.$$

Let U be obtained from A by deleting an element from each bad r-vertex subset. So, U has no bad r-vertex subsets. Also,

$$\mathbb{E}|U| \;\geq\; \mathbb{E}|A| - \mathbb{E}[\text{the number of bad } r\text{-vertex subsets of } A] \;\geq\; n\alpha^t - \binom{n}{r}\Big(\frac{m}{n}\Big)^t.$$

Thus, there exists some U with at least this size, with the property that all its r-vertex subsets have more than m common neighbors. □